

Underdetermined Systems - Requirements and Restrictions

Computing resources, software, experimental controls and limitations for a reproducible run

1. Purpose

This document records the conditions that matter when reproducing the large underdetermined experiments. Some conditions define the mathematical experiment and should remain fixed. Others describe the computing environment and may need to be adapted to current University rules. A change to the computing environment is not automatically a change to the experiment, but it should be recorded.

2. Problem dimensions

The benchmark must remain underdetermined, meaning n is greater than m . The main experiments use $m = 5000$ and $n = 15000$. The matrix-size study also uses $n = 20000$ and $n = 40000$.

3. Scientific parameters

Parameter	Recorded values
m	5000
n	15000 baseline; 15000, 20000, 40000 in matrix-size study
Condition parameter	10, 100, 1000, 10000, 20000
Sparsity	0.90, 0.95, 0.98, 0.99, 0.999
Lambda	0, 1e-8, 1e-6, 1e-4, 1e-2, 1e-1, 1
Sparsity pattern	random, banded, block
Spectrum	linear, exponential, polynomial, clustered
Repeats	10 per configuration
Noise level	0.01
Initial random seed	42; run number changes the seed for repeated problems
CG tolerance	$rtol = 1e-8$, $atol = 0$, $maxiter = 5000$
LSQR tolerance	$atol = 1e-8$, $btol = 1e-8$, $iter_lim = 5000$
LSMR tolerance	$atol = 1e-8$, $btol = 1e-8$, $maxiter = 5000$

4. The six studies are controlled one-factor experiments

The code does not take every combination of every listed parameter. Each study changes one factor while fixing the other parameters to the values defined for that study. This structure should not be replaced by a full Cartesian parameter sweep if the aim is to reproduce the dissertation experiment.

5. Matrix construction

- Use the supplied sparse matrix construction rather than replacing it with a dense matrix.
- Do not reduce m or n merely to make the calculation finish on a laptop or low-memory machine.
- Do not change the sparsity pattern while retaining the same experimental label.
- The condition parameter is the control used by the generator; it should not be described as a guaranteed measured condition number of every realised matrix.

6. Solver implementation restrictions

- Keep CG, LSQR and LSMR as the named iterative methods.
- The supplied CG implementation applies $A^T A$ through a LinearOperator; do not replace this with an explicitly formed normal matrix when reproducing that implementation.
- LSQR and LSMR use the supplied damping formulation, with damp equal to the square root of λ .
- Do not silently remove non-zero regularisation.
- Do not change convergence tolerances or iteration limits and then compare the result directly with the dissertation timings.
- Do not treat Direct as a successful large-scale comparison method.

7. Direct solver: important practical limitation

The Direct function in the supplied large-system code explicitly forms $A^T A$ and then solves the resulting sparse system. This is fundamentally different from the matrix-free CG operation and has a much heavier memory and factorisation burden.

The dissertation records that Direct did not successfully complete the large underdetermined experiments. This is not simply a case of using slightly too little memory. For the large dimensions in this section, Direct should be regarded as a reference attempt that is expected to be impractical in many cases.

A 128 GB allocation is the practical minimum reference for attempting the Direct cases. An allocation below 128 GB should not be used to judge whether Direct is viable for the benchmark. At the same time, allocating 128 GB or substantially more does not guarantee completion. The direct factorisation can still fail or become impractical because of memory growth, sparse fill-in and computational cost.

For this reason, a reproduction in which CG, LSQR and LSMR complete while Direct does not is consistent with the recorded experiment. The successful numerical comparisons in the dissertation are the iterative methods.

8. Software requirements

Component	Requirement
Python	Compatible Python 3 environment on CSF.
NumPy	Array operations and random-number generation.
SciPy	Sparse matrices and CG, LSQR, LSMR and sparse direct solve.
pandas	CSV handling and result aggregation.
Matplotlib	Plot generation.
Environment	The original work used an Anaconda-based environment; exact versions should be recorded.

9. CSF access

CSF access, account setup and institutional authentication are prerequisites. The current University procedure must be followed because access arrangements can change. This repository does not attempt to document the login process.

10. Memory allocation

The reference workflow used 128 GB of memory. This is especially important for the Direct test. Allocations below 128 GB are expected to fail for the Direct cases and should not be used to assess Direct viability.

For CG, LSQR and LSMR, the memory requirement is substantially lower than for Direct, but the actual allocation still has to be large enough for the selected matrix dimensions, sparse data structures and the concurrent work performed by the task.

11. Wall time

The recorded workflow used a 24-hour Slurm allocation. This is the reference wall-time setting for reproduction. Current CSF limits must be checked before submitting a new job.

Wall time and memory solve different problems. Increasing the wall-time limit does not fix an out-of-memory failure, and increasing memory does not guarantee that a direct factorisation will complete.

12. CPU allocation is user-dependent

The number of CPUs per task must not be treated as a fixed historical requirement. It depends on the current University allocation, account or project limits, and the CSF partition or queue.

- Check the current University CSF documentation before submission.
- Use a CPU count that is actually permitted for the account and partition.
- Record the CPU count used for the reproduction.
- Report the CPU allocation with timing results if execution times are compared.

13. Reference Slurm structure

The recorded workload has six experiment studies and four solver choices, giving 24 experiment-solver combinations. A 24-element Slurm array is therefore a convenient way of organising the workload.

```
# Reference structure only
#SBATCH --time=24:00:00
#SBATCH --mem=128G
#SBATCH --cpus-per-task=<CPU_COUNT>
#SBATCH --array=1-24
```

The CPU placeholder is deliberate. It must be replaced by a value allowed by the current University environment. A historical job number is not part of the reproducibility specification.

14. Output and repository restrictions

- The source repository can contain the two Python files and the three explanatory PDFs for this section.
- Generated CSV files do not need to be permanently stored if the supplied code can regenerate them.
- Generated PNG files do not need to be permanently stored if the plotting program can regenerate them.
- If generated outputs are retained for comparison, identify them clearly as generated results.
- Do not manually edit a generated CSV and then present it as a direct benchmark result.

15. What should be recorded for a serious reproduction

- Repository version or commit used.
- Date and time of execution.
- Python version.
- NumPy, SciPy, pandas and Matplotlib versions.
- CSF partition or queue.
- Memory requested and, where available, memory actually used.
- CPU count requested and relevant hardware information.
- Wall-time limit.

- Any change required by current CSF policy.
- Locations of the generated numerical CSV files and plots.

16. Interpreting failure correctly

A failed job is not automatically a software error. The first distinction should be between an environment problem, a Slurm/resource problem, a numerical convergence problem and the known Direct-solver limitation.

For Direct, failure at the large underdetermined scale is consistent with the dissertation. It should not be 'fixed' by changing the matrix dimensions, removing regularisation, changing the solver formulation or replacing the direct method with an iterative method while still reporting the result as Direct.

17. Final reproduction checklist

- The current CSF access procedure has been followed.
- Underdetermined.py and Underdetermined_Plots.py are the supplied repository versions.
- The six studies and their parameter values have been checked.
- Ten repeats, noise level 0.01 and initial seed 42 are preserved.
- The solver tolerances and maximum iteration limits are preserved.
- A 24-hour allocation is used where current CSF rules permit it.
- At least 128 GB is requested if Direct is attempted.
- The Direct result is interpreted as a reference test rather than an expected successful large-scale calculation.
- The CPU allocation is selected according to current University limits and recorded.
- The numerical stage is completed before the plotting stage.
- Generated CSV and PNG files are treated as reproducible outputs rather than permanent source dependencies.